MAESTRO INFRASTRUCTURE SETUP IN AWS

This document provides the information on the configuration of an instance in AWS Cloud for on premise installation of Maestro.

The described setup is necessary to enable necessary access and performance for Maestro application.

The infrastructure setup in AWS includes several steps:

- Maestro Instance Setup
- Certificate Validation
- OnPrem Infrastructure Deployment

1 MAESTRO INSTANCE SETUP

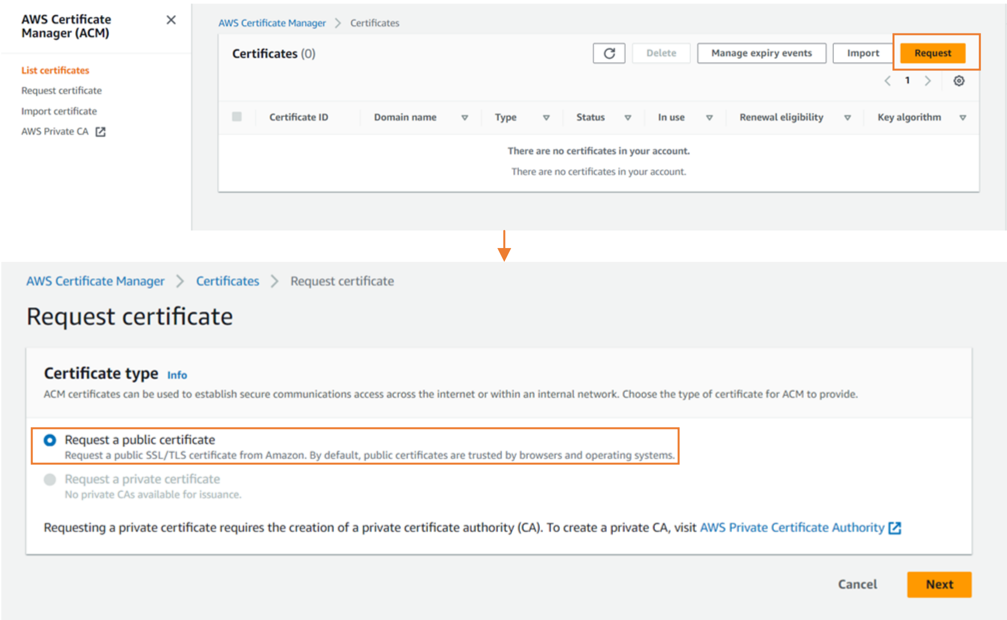
1.1 Maestro Instance Configuration

- **Minimum instance type:** m7a.large (2 vCPU, 8 GB RAM)
- **Disk size:** 100 GB
- **Operating System:** Ubuntu 22.04

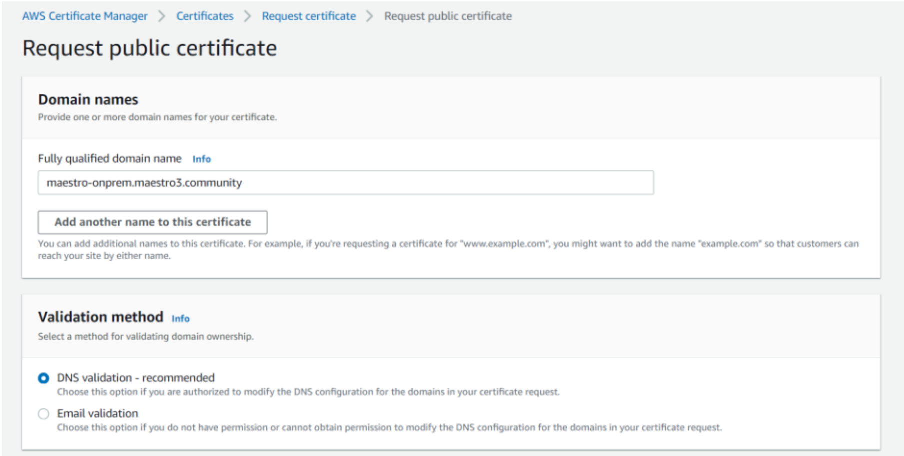
1.2 Maestro Instance Setup

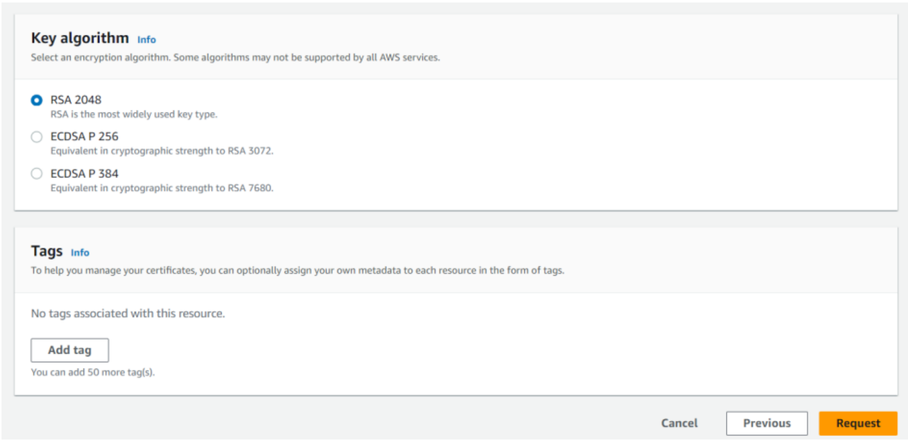
To setup an instance where Maestro will be installed, complete the following steps:

1. Navigate to **AWS Certificate Manager**, click **Request** and select **Request a public certificate**. This certificate needed to enable https access for Maestro.

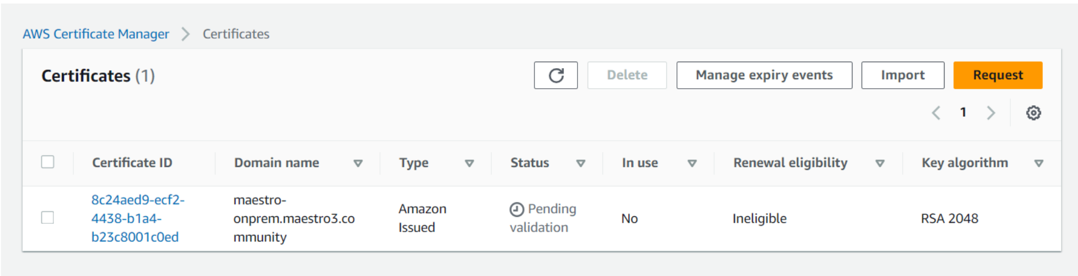


2. Specify a fully qualified domain name and request certificate as shown below:

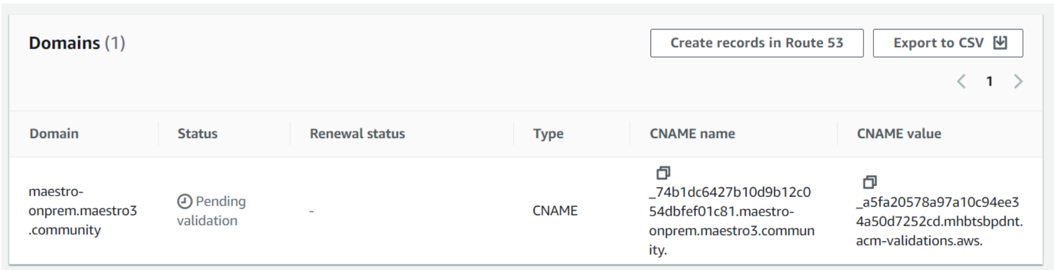




3. Once the certificate is requested it will be displayed as pending validation. Find it in the Certificates list:



4. Open the created certificate entity and scroll down to the Domains section:

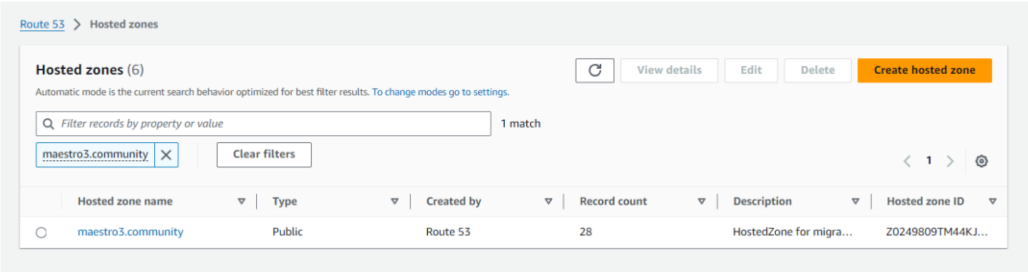


5. Copy “CNAME name” and “CNAME value” – it will be needed in following step to validate certificate.

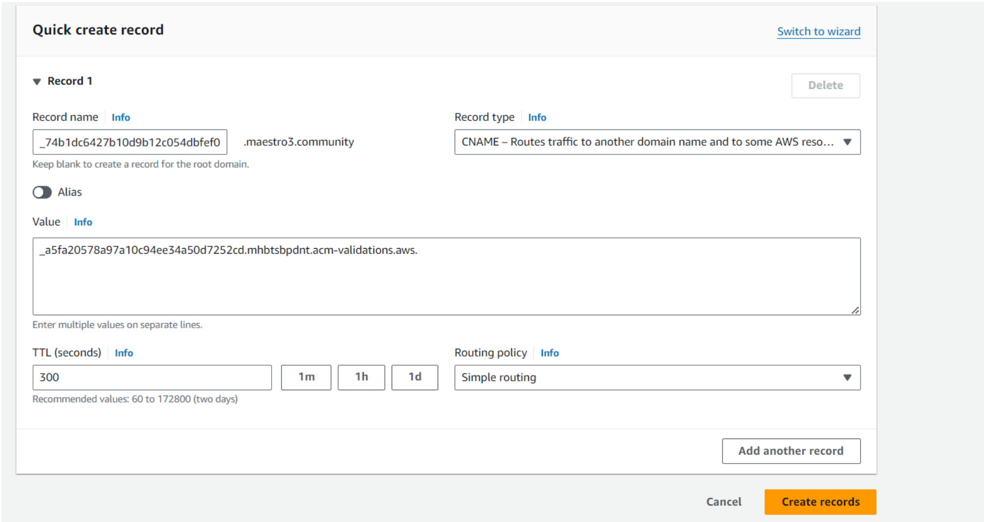
2 CERTIFICATE VALIDATION

In order to validate a certificate, follow the steps below:

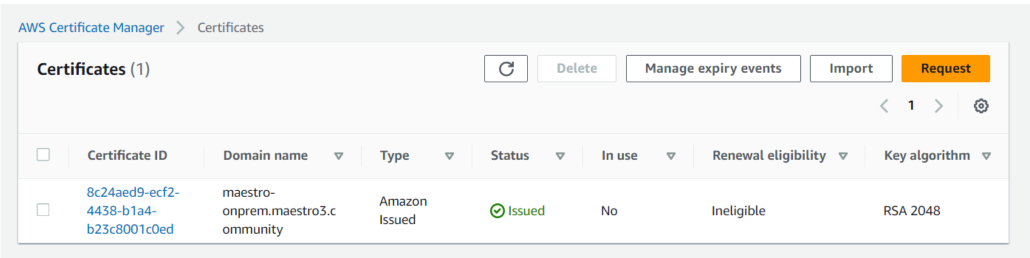
1. Navigate to **Route53**, select **Hosted Zones** and select a hosted zone related to fully qualified domain name specified during certificate request



2. Create a record to verify the certificate (Record name and Value can be obtained from step 5 of Maestro Instance Setup section – these are “CNAME name” and “CNAME value” respectively)



When the record is created, validation can take some time. Once the validation is completed, the certificate status will be changed to “Issued”:



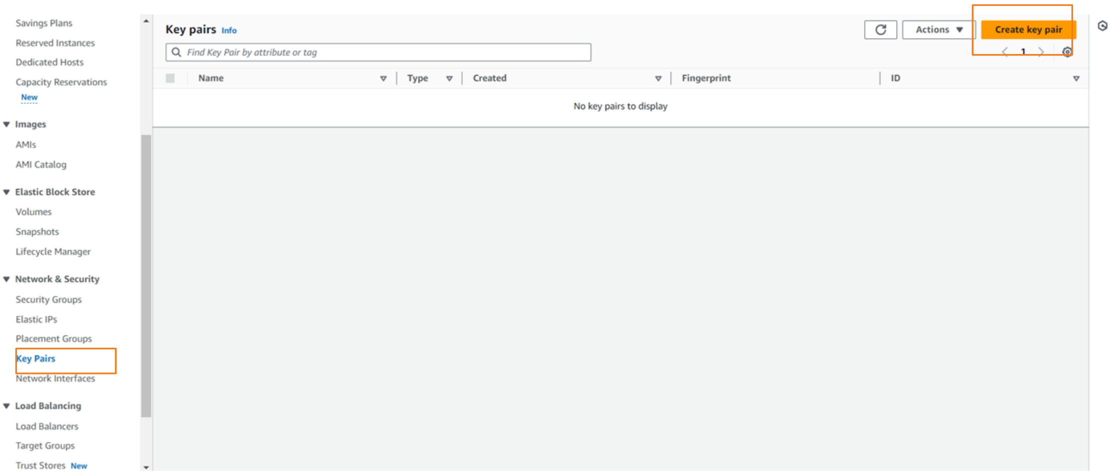
3 DEPLOY MAESTRO ONPREM INFRASTRUCTURE

When installed in AWS, Maestro also needs additional infrastructure. The deployment is performed according to the steps listed below in this section.

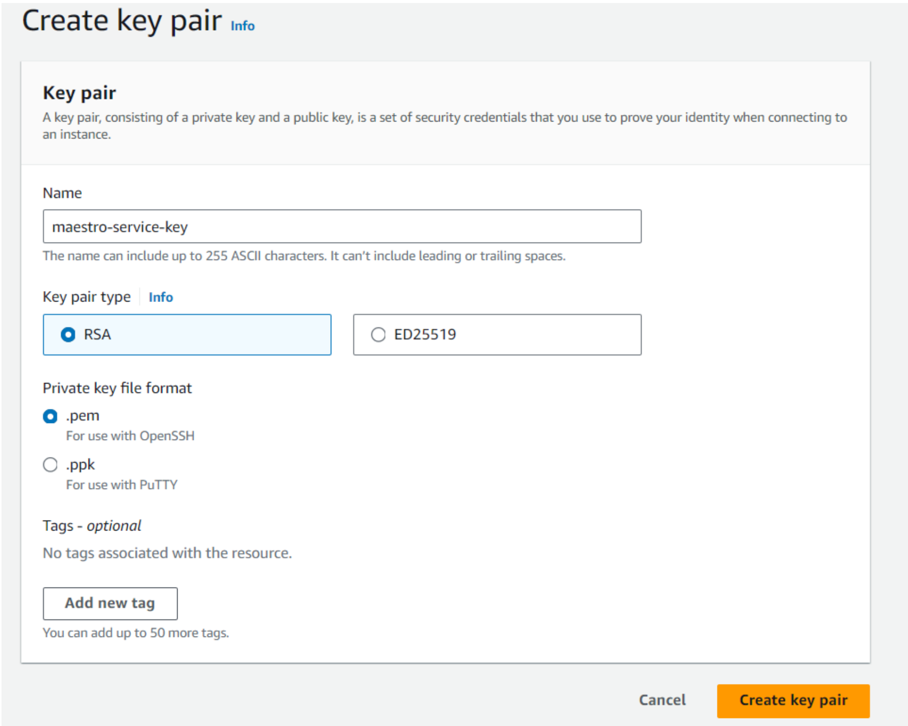
3.1 Create a Key Pair

To create a Key Pair to be used to connect to the instance, follow the steps below:

1. Navigate to **EC2** -> **Key Pairs** -> **Create Key Pair**



2. Set the key name as “maestro-service-key” and change Private key file format to “.pem”. After creating key will be downloaded.

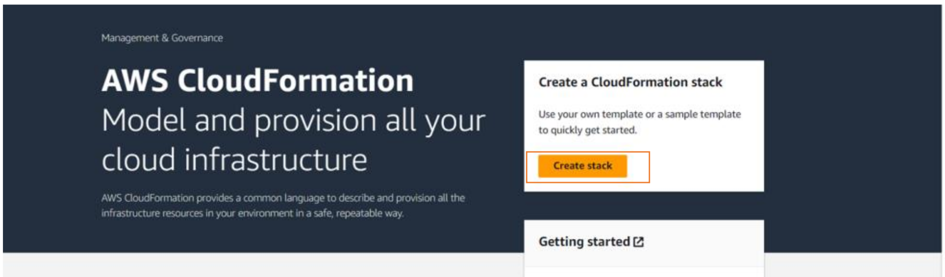


3.2 Launch AWS CloudFormation template

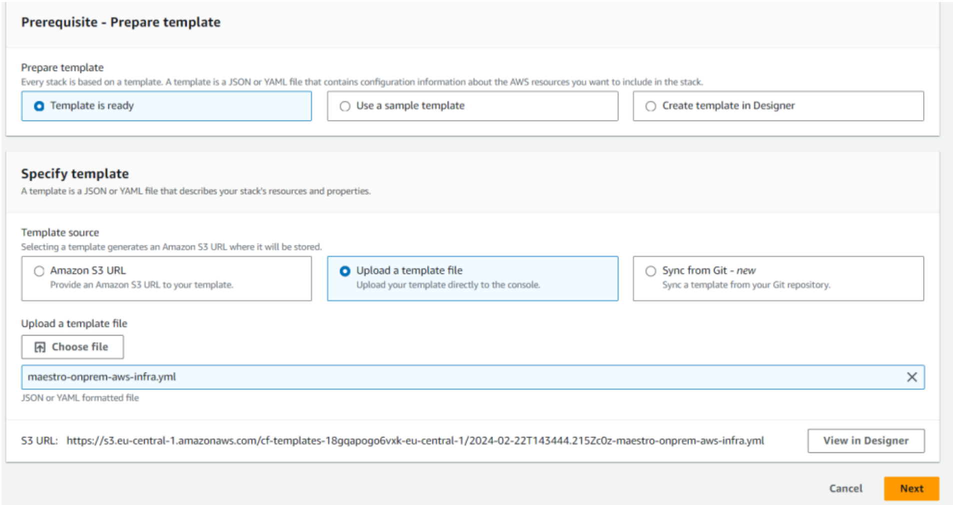
The necessary infrastructure is deployed via an AWS CloudFormation template. To do this, complete the following steps:

Please use the template provided together with this instructions.

1. Go to **CloudFormation** -> **Create Stack**



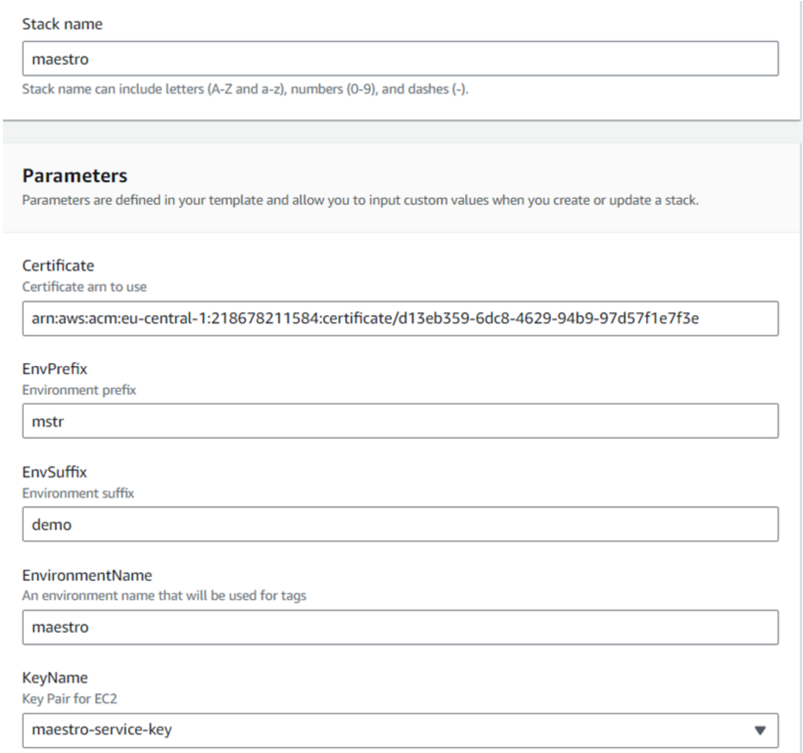
2. Choose "**Upload a template file**" as template source and upload the file provided with this instruction.



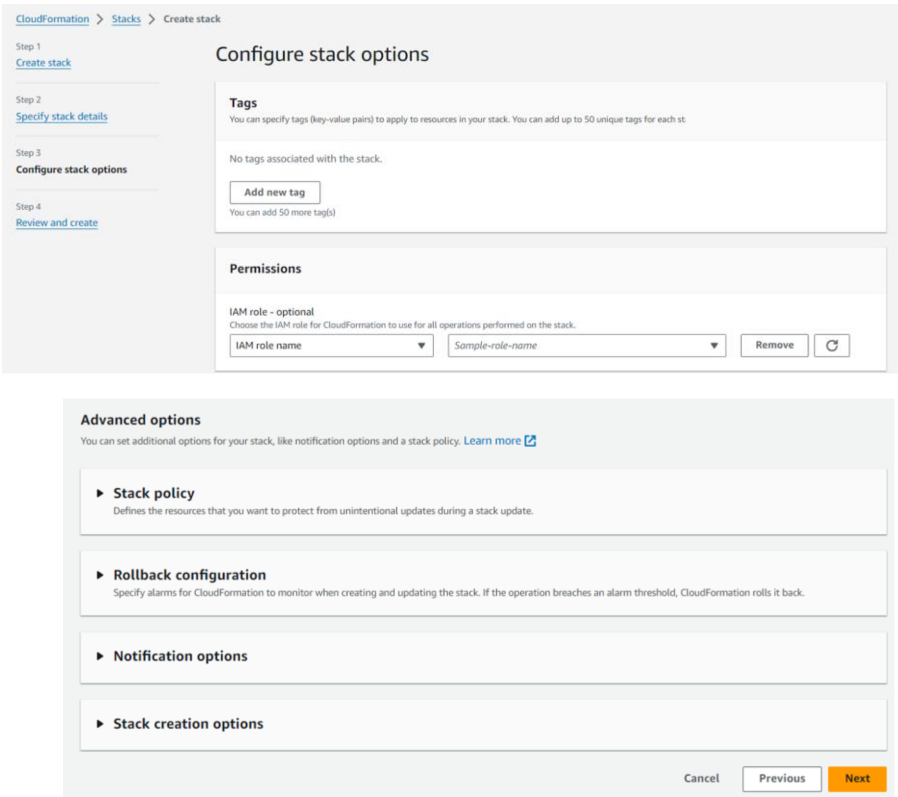
3. Specify the following parameters for the template launch:

- Stack name: maestro
- Certificate: ACM certificate ARN created in previous step. To obtain it, navigate to AWS Certificate Manager, open created certificate and copy ARN

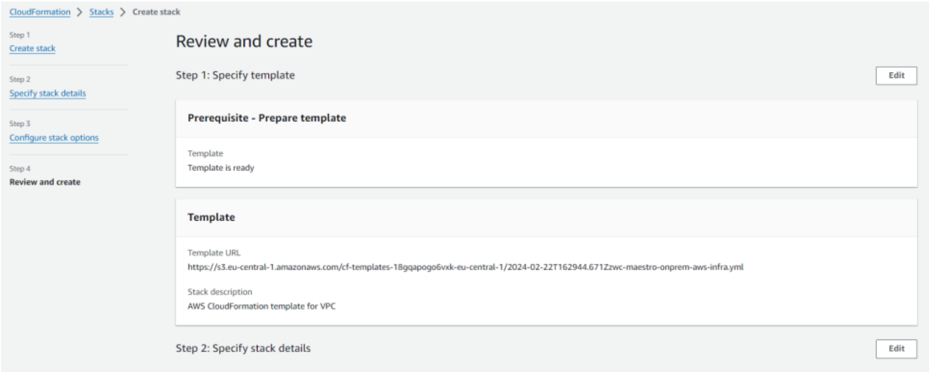
- KeyName: The key created on step 3.1

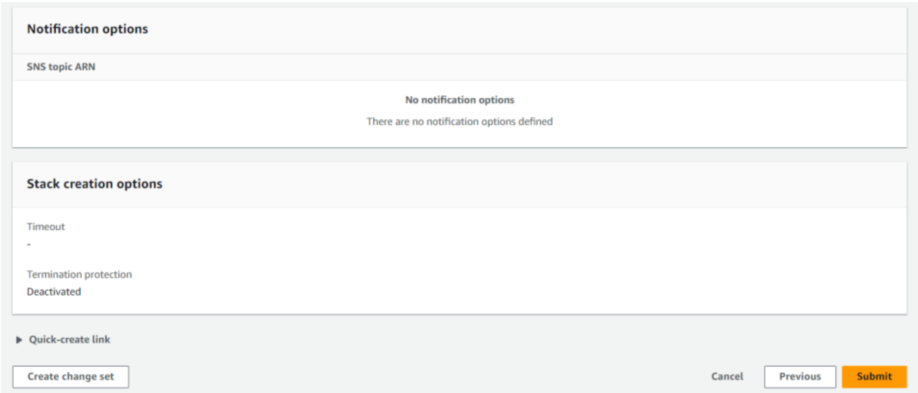


4. Leave other parameters default, scroll down, and click Next. At "Configure stack options" section keep everything default and move to the **Next** section.

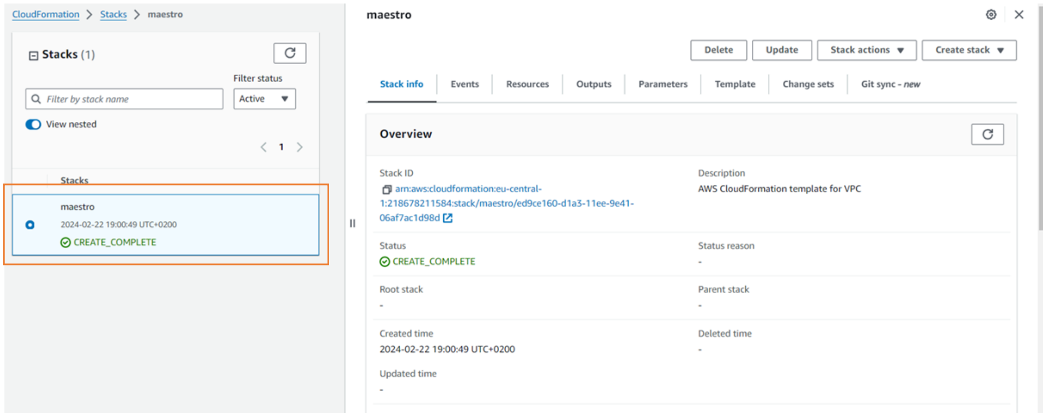


5. Scroll down and press Submit





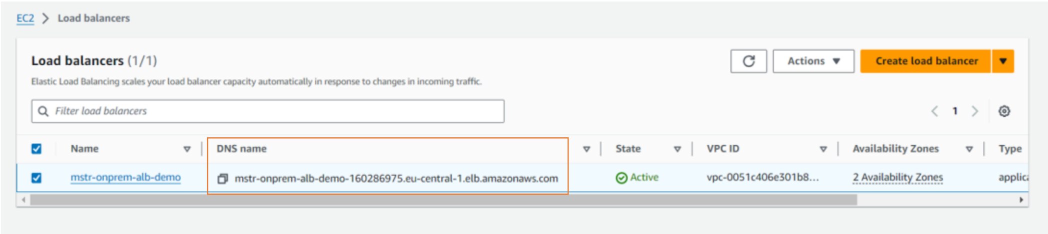
Infrastructure creation may take some time. After all resources are created, the stack will be in CREATE_COMPLETE state.



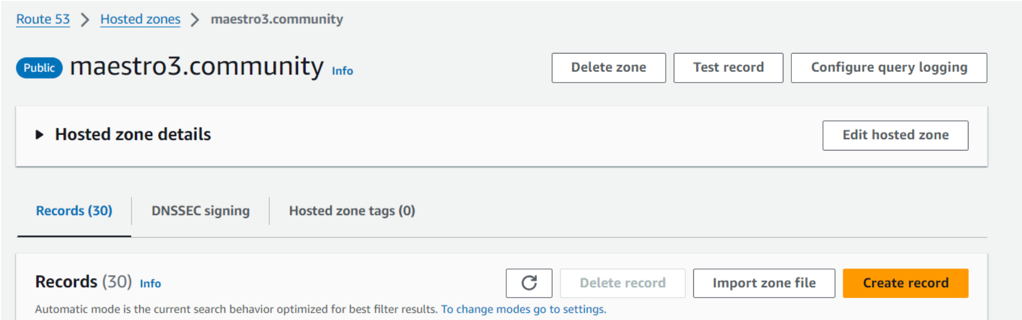
3.3 Create Route53 record for Load Balancer

To create a Route53 record for Load Balancer, do the following:

1. Navigate to **EC2** -> **Load Balancer**. Retrieve Load Balancer DNS name. it will be used in following step.



2. Navigate to **Route53** -> **Hosted Zones**. Select Hosted zone used for certificate creation and press Create record:



3. Use your certificate from earlier steps as Record name. Choose "CNAME" for Record type. In value field paste Load Balancer DNS retrieved from previous step and click Create record.